

AP CALCULUS AB – UNIT 5

5.10 Optimization Problems



Name: _____ Class: _____ Date: _____

Lesson Overview and Learning Objectives

- Identify an objective quantity and a constraint relating variables.
- Rewrite the objective as a one-variable function on a meaningful domain.
- Find and justify an absolute maximum or minimum using critical points and endpoints.
- Interpret optimized dimensions and values with correct units.

Keys: Essential Knowledge and Vocabulary

Objective	The quantity to maximize or minimize, such as area, volume, cost, distance, or time.
Constraint	The equation linking the variables and fixed data.
Feasible domain	Input values allowed by geometry, physics, and the context.
Candidate set	Interior critical points and relevant endpoints or limiting boundary cases.
Verification	Use sign analysis, second derivative, or comparison of objective values to prove optimality.
Interpretation	Report dimensions and the optimized quantity, including units and appropriate rounding.

Explanation, Strategy, and Common Errors

- Draw and label a model, then state the objective and constraint separately.
- Use the constraint to eliminate variables until the objective depends on one variable.
- Write the physically meaningful domain before differentiating.
- Find critical points and compare all feasible candidates; do not stop after solving $F' = 0$.
- Answer the original question, not merely the value of the chosen variable.

Solved Example: Maximum rectangle

A rectangle has perimeter 40 m. Find dimensions of maximum area.

Answer and Reasoning

$2x + 2y = 40$, so $y = 20 - x$. $A = x(20 - x)$ on $0 < x < 20$. $A' = 20 - 2x = 0$ gives $x = 10$, $y = 10$; the maximum area is 100 m^2 .

Solved Example: Open-top box

Squares of side x are cut from a 20 cm by 14 cm sheet. Maximize the box volume.

Answer and Reasoning

$V = x(20 - 2x)(14 - 2x)$ on $0 < x < 7$. $V' = 12x^2 - 136x + 280 = 4(3x^2 - 34x + 70)$. The feasible critical point is $x = (17 - \sqrt{79})/3 \approx 2.704$ cm; compare endpoints to verify the maximum.

Solved Example: Minimum material cylinder

A closed cylinder has volume 500π cm³. Express surface area in one variable.

Answer and Reasoning

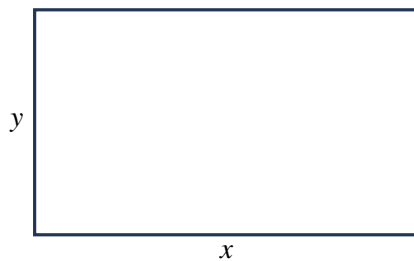
$\pi r^2 h = 500\pi$, so $h = 500/r^2$. Then $S(r) = 2\pi r^2 + 2\pi r(500/r^2) = 2\pi r^2 + 1000\pi/r$, $r > 0$.

Solved Example: Closest point

Find the point on $y = x^2$ closest to $(0, 3)$.

Answer and Reasoning

Minimize squared distance $D^2 = x^2 + (x^2 - 3)^2$. Derivative $2x + 4x(x^2 - 3) = 2x(2x^2 - 5)$. Candidates $x = 0, \pm\sqrt{5/2}$; comparison gives closest points $(\pm\sqrt{5/2}, 5/2)$.

Shared Representation / Data

Objective: $A = xy$
 Constraint: $2x + 2y = P$
 Domain: $0 < x < P/2$

AP Exam Language

For theorem applications, state every hypothesis. For graph analysis, justify sign changes and compare all required candidates. For optimization, state the objective, constraint, feasible domain, and verification.

Leveled Practice**Easy Foundations**

Build fluency with the definition, notation, and one-step applications.

1. For a rectangle with fixed perimeter P , express y in terms of x .

2. State the feasible domain for rectangle width x in the previous problem.

3. If an objective is continuous on $[a, b]$, which candidates are checked?

4. Why is minimizing distance equivalent to minimizing squared distance?

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. A rectangle has area 96 m^2 . Express perimeter as a function of width x .

6. Find dimensions minimizing that perimeter.

7. An open-top square-base box has volume 108 cm^3 . Express material area in terms of base side x .

8. A farmer fences three sides of a rectangle against a river using 120 m. Express area.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. Find the maximum area in the river-fencing problem.

10. A closed cylinder has fixed volume V . Show minimum surface area occurs when $h = 2r$.

11. Find the point on $y = 2x + 1$ closest to the origin.

12. A poster has printed area 384 cm^2 , side margins 2 cm and top/bottom margins 3 cm. Express total area using printed width x .

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. Optimize the poster dimensions from the previous problem.

14. A right triangle has hypotenuse 10. Maximize its area.

15. A cylindrical can must hold 1000 cm^3 and has no top. Find the radius minimizing material.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. A rectangle is inscribed under $y = 12 - x^2$ above the x -axis, symmetric about the y -axis. Maximize area.

17. Prove among rectangles with fixed area, the square has minimum diagonal.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. The objective function is

- (A) the fixed equation only
- (B) the quantity to maximize or minimize
- (C) the derivative
- (D) the domain

2. After finding $F'(x) = 0$, the next essential step is

- (A) stop
- (B) ignore endpoints
- (C) verify and compare feasible candidates
- (D) differentiate the constraint again

3. A rectangle with perimeter 24 has maximum area when dimensions are

- (A) 1 by 11
- (B) 4 by 8
- (C) 6 by 6
- (D) 5 by 7

4. A physically meaningful domain is needed because

- (A) derivatives require integers
- (B) not every algebraic value represents a possible configuration
- (C) functions cannot be negative
- (D) endpoints are never allowed

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. A rectangle has area 150 and minimum perimeter. Each side is closest to

- (A) 10.00
- (B) 12.25
- (C) 15.00
- (D) 18.37

6. A square-base open box of volume 256 has minimum material when base side x is closest to

- (A) 5.04
- (B) 6.35
- (C) 8.00
- (D) 10.08

AP-Style Free-Response Practice

No Calculator

A rectangular enclosure is divided into three equal pens by two fences parallel to its width. A total of 240 m of fencing is available.

(a) Let x be the width crossed by four fence segments and y the length crossed by two. Write the constraint and area. [2 points]

(b) Express area in terms of x and find the critical point. [3 points]

(c) Give the optimal dimensions and justify the maximum. [2 points]

Calculator Permitted

An open-top box is made by cutting equal squares from the corners of a 24 cm by 18 cm sheet and folding the sides.

(a) Write the volume function and domain. [2 points]

(b) Use a calculator to find the feasible critical point that maximizes volume. [3 points]

(c) Find the maximum volume to the nearest cubic centimeter. [2 points]